Sentiment Analysis — User Manual

Cross-lingual sentiment analysis for Dravidian languages (Tamil, Malayalam, Kannada, Telugu)

Application URL: `http://localhost:8080`

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1. Application Overview

The Sentiment Analysis application determines whether a piece of text expresses a **Positive**, **Negative**, or **Neutral** sentiment. It is powered by the **XLM-RoBERTa** multilingual transformer model, trained specifically for Dravidian languages, and works with text written in native script **or** in romanized (English-letter) form.

Multi-language Support

Tamil, Malayalam, Kannada, and Telugu in both native script and romanized Latin text.

Auto Transliteration

Romanized text is automatically converted to native script via ICU4J for higher accuracy.

Confidence Score

Every result shows a confidence percentage and per-class probability breakdown.

Analysis History

All your analyses are saved and can be browsed, filtered by language, and paginated.

2. Model Information

The application uses a pre-trained multilingual transformer model, exported to ONNX format for fast inference inside the JVM.

Sentiment Model Details

MODEL NAME	`cardiffnlp/twitter-xlm-roberta-base-sentiment`
ARCHITECTURE	XLMRobertaForSequenceClassification
BASE MODEL	XLM-RoBERTa Base (multilingual, 100+ languages)
HIDDEN LAYERS	12 transformer layers, 768 hidden units
OUTPUT LABELS	Negative · Neutral · Positive
INFERENCE ENGINE	DJL (Deep Java Library) 0.36.0 + ONNX Runtime
TRANSLITERATION	ICU4J 75.1 (Latin → Tamil / Telugu / Kannada / Malayalam)
MAX INPUT LENGTH	256 tokens
LOW CONFIDENCE THRESHOLD	60% (results below this are flagged)
MODEL PATH	`./models/xlmr-sentiment`

ⓘ The model is originally from **Hugging Face** (`cardiffnlp/twitter-xlm-roberta-base-sentiment`), fine-tuned on multilingual Twitter data. It is exported to ONNX format via the `python/export_model.py` script and runs entirely inside the JVM — no Python or GPU is needed at runtime.

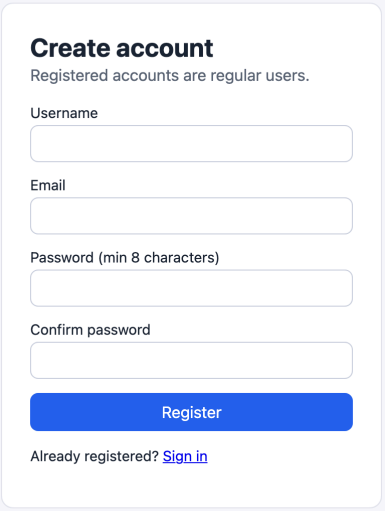
3. Getting Started

Before using the application, ensure the following prerequisites are met:

REQUIREMENT	DETAILS
Web browser	Any modern browser (Chrome, Safari, Firefox, Edge)
Application URL	`http://localhost:8080` (default, or the URL provided by your administrator)
Account	Register a new account or use credentials provided to you

ⓘ The application uses role-based access. Regular users access the **User area** (`/user/*`). Administrators access the **Admin area** (`/admin/*`). Each role can only see its own pages.

4. Registration



Registration page — `/register`

1 Open the Registration page

Navigate to `http://localhost:8080/register` or click "Register" on the login page.

2 Fill in your details

Username

— must be unique, max 30 characters.

Email

— must be a valid, unique email address.

Password

— minimum 8 characters.

Confirm password

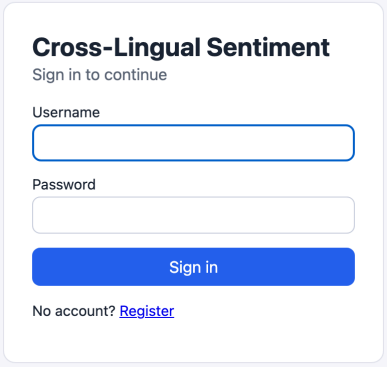
— must match the password.

3 **Click "Register"**

On success, you are redirected to the login page with a confirmation message.

⚠ If you see a validation error (red text below a field), correct the highlighted field and try again. Common issues: username already taken, email already registered, or passwords don't match.

5. Login



Login page — `/login`

1 **Open the Login page**

Navigate to `http://localhost:8080/login`. Any unauthenticated URL automatically redirects here.

2 **Enter your credentials**

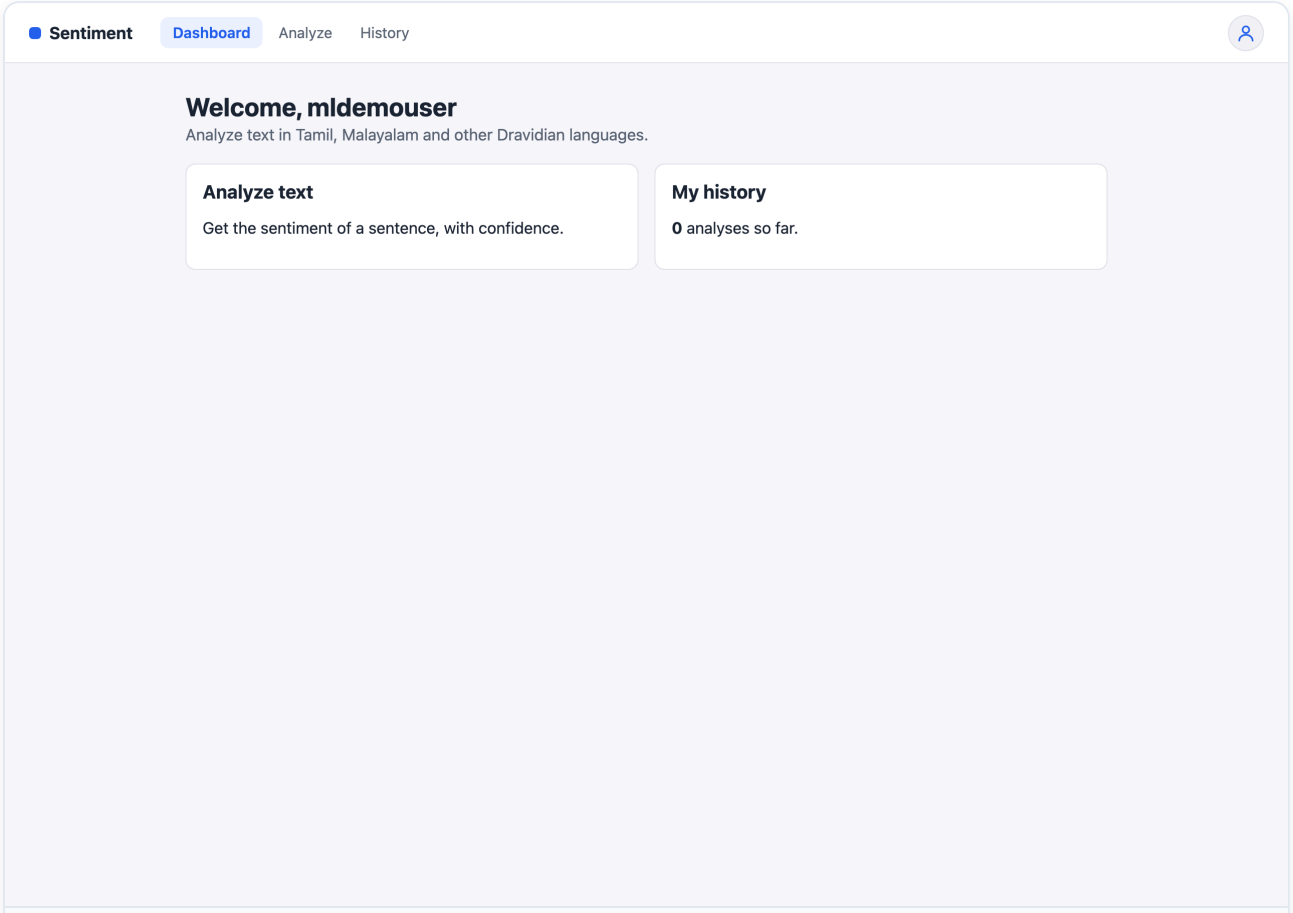
Type the username and password you registered with.

3 **Click "Sign in"**

On success, you are redirected to your **User Dashboard**.

✓ After successful registration, you'll see a green banner: *"Account created. Please sign in."*

6. User Dashboard



User Dashboard — `/user/dashboard`

The dashboard is your landing page after login. It provides quick links to:

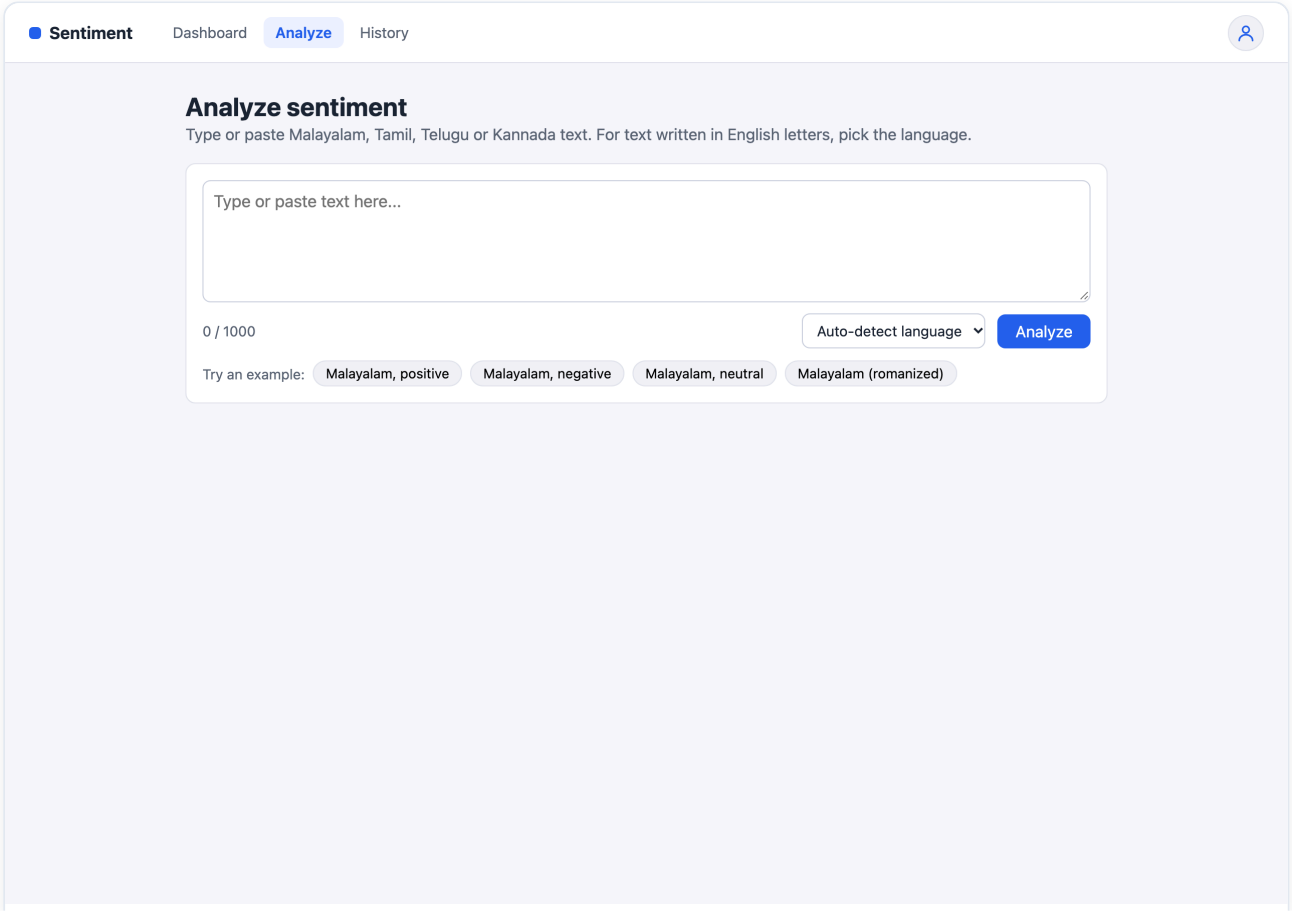
- **Analyze text** — opens the Analyze page to run sentiment analysis
- **My history** — shows how many analyses you've run and links to the full History page

Navigation Bar

The top navigation bar is visible on every user page and includes:

ITEM	ACTION
Sentiment (brand)	Returns to the Dashboard
Dashboard	Opens the Dashboard
Analyze	Opens the Analyze page
History	Opens the History page
Avatar icon (top-right)	Shows your username and a **Logout** button

7. Analyzing Sentiment



Analyze page — `/user/analyze`

1 **Navigate to the Analyze page**

Click

Analyze

in the nav bar, or click the "Analyze text" tile on the Dashboard.

2 **Enter your text**

Type or paste text into the text area (up to 1,000 characters). The character counter updates in real time.

3 **Select a language (if needed)**

- **Native script**

(e.g., മലയാളം): leave the dropdown on *"Auto-detect language"*.

- **Romanized text**

(e.g., "Ee cinema valare nallathanu"): select the correct language (Tamil, Malayalam, Kannada, or Telugu) from the dropdown. This enables automatic transliteration.

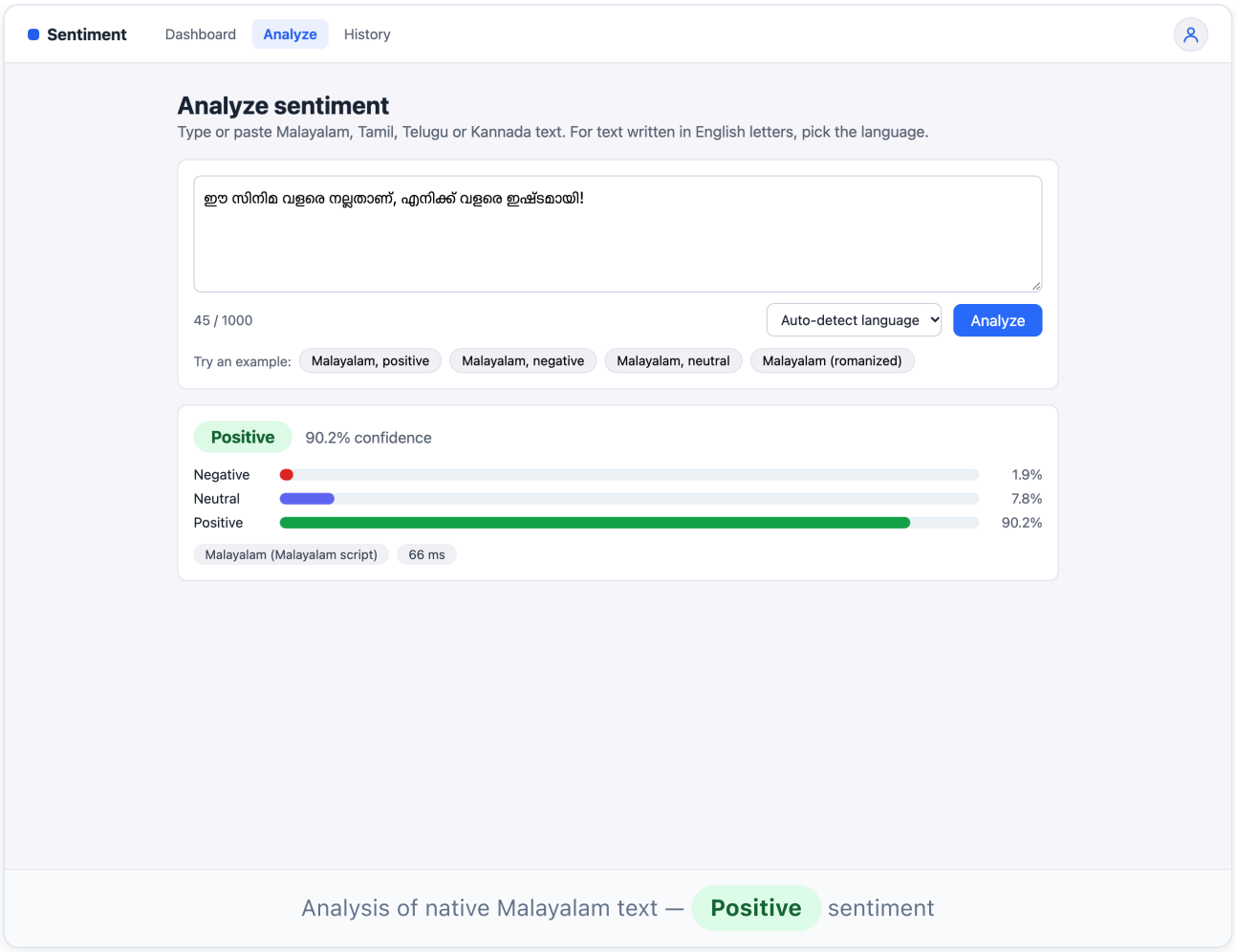
4 **Click "Analyze" (or press `Ctrl` + `Enter`)**

The button changes to "Analyzing..." while processing.

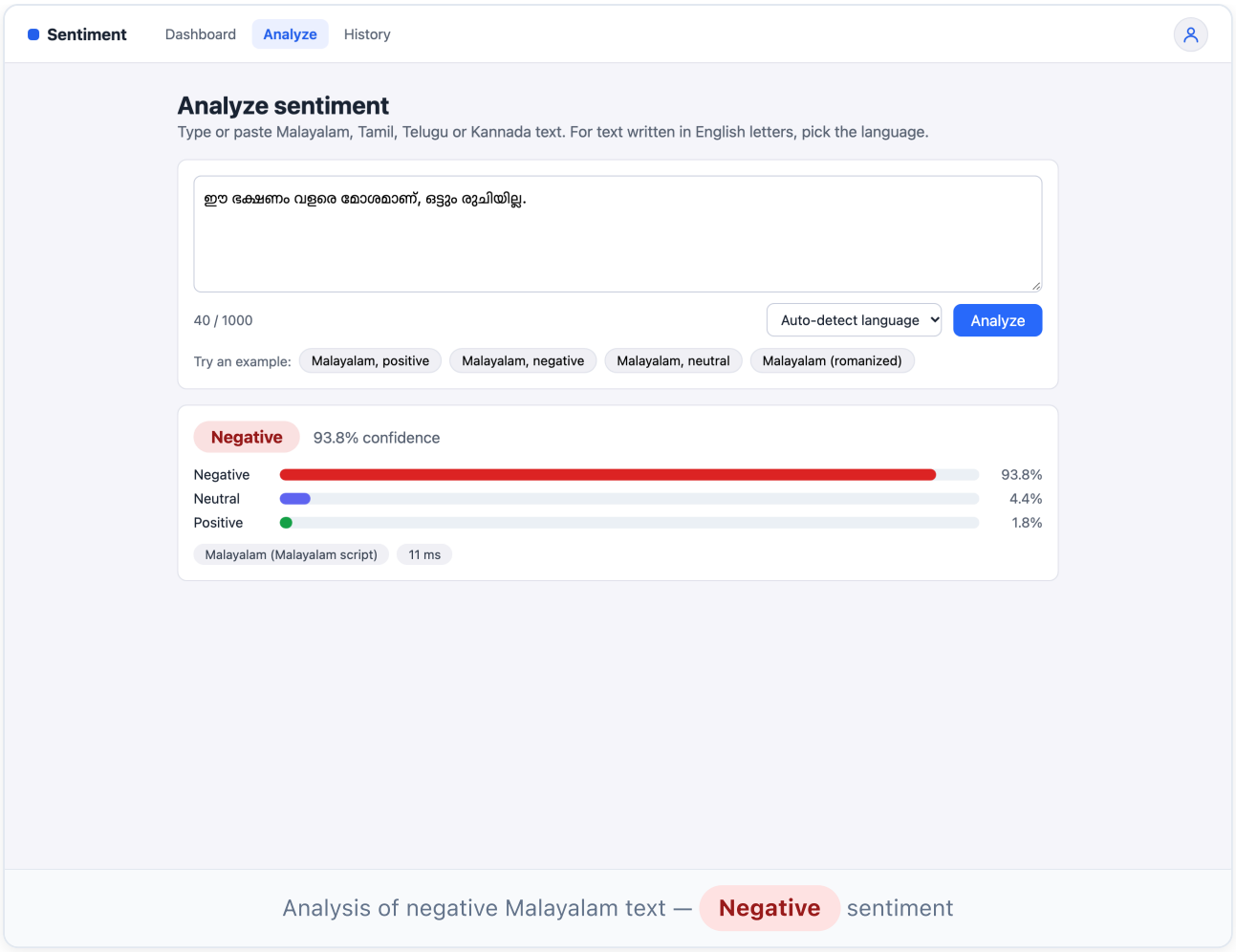
5 **View the result**

The result panel appears below with the sentiment label, confidence score, probability bars, detected language, and latency.

Native Script Analysis



Negative Sentiment Example

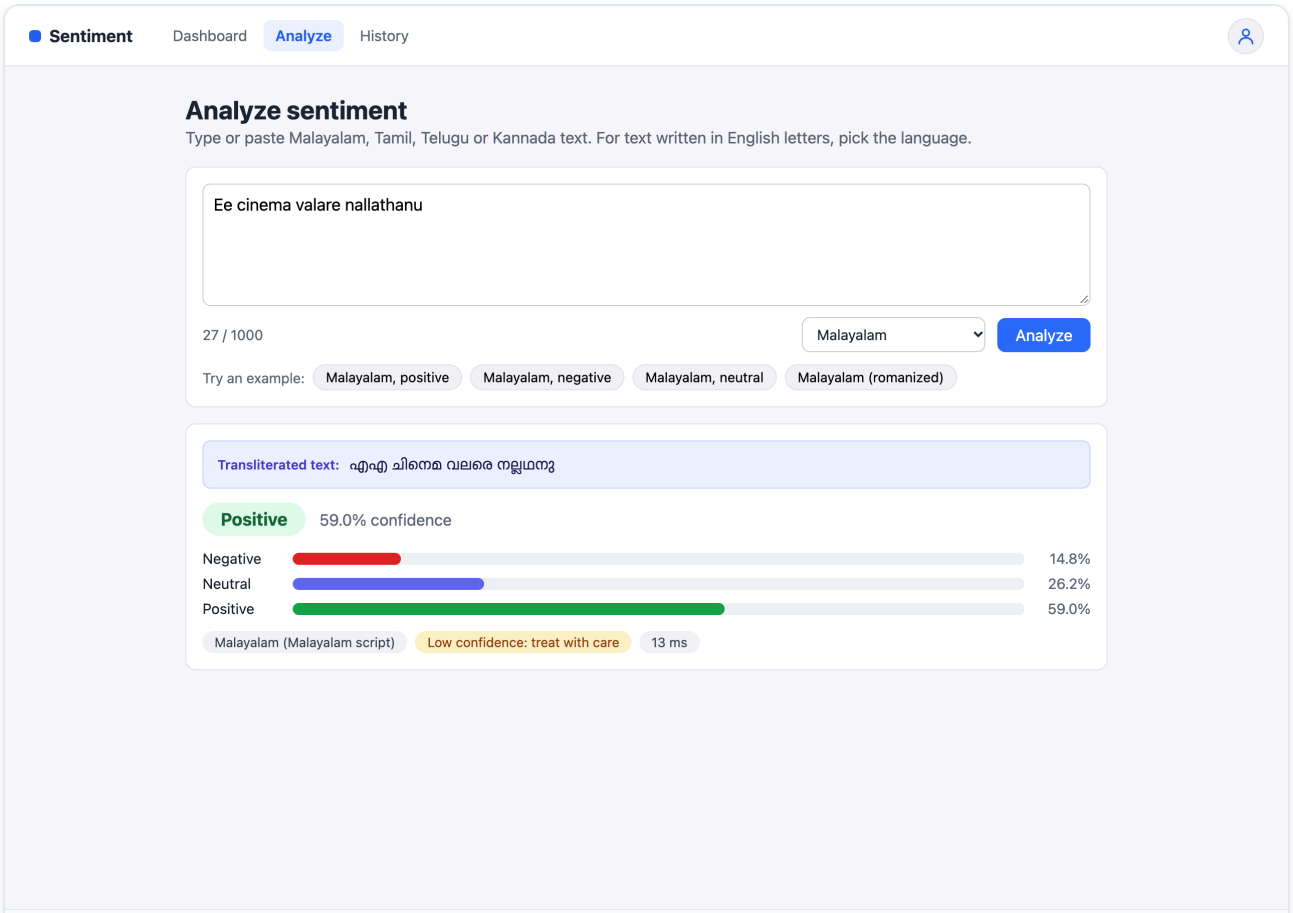


Example Chips

Below the text area, pre-loaded example buttons let you quickly test the system. Clicking a chip fills the text area and (for romanized examples) selects the language automatically.

8. Transliteration (Romanized Text)

When you type Dravidian text using English letters (romanized / Latin script), the model's accuracy drops significantly (from ~100% to ~20%). The application solves this by **automatically transliterating** your romanized text into native script before analysis using the **ICU4J** library.



Romanized text "Ee cinema valare nallathanu" transliterated to Malayalam script — **Positive** sentiment

How it works

1 **You type romanized text**

Example: `Ee cinema valare nallathanu`

2 **You select the language**

Choose "Malayalam" from the language dropdown (required for romanized text).

3 **The system detects Latin script**

It recognizes the input is in English letters, not native script.

4 **Automatic transliteration**

The text is converted to native Malayalam script (shown in the blue box above the result).

5 **Analysis runs on native script**

The XLM-RoBERTa model processes the transliterated text, producing much higher accuracy.

ⓘ The transliterated text is displayed in a blue box above the result so you can verify the conversion. If you're already typing in native script, no transliteration occurs.

⚠ **Important:** You must select the correct language from the dropdown when entering romanized text. Without it, the system cannot determine which script to transliterate into, and accuracy will be low (~20%).

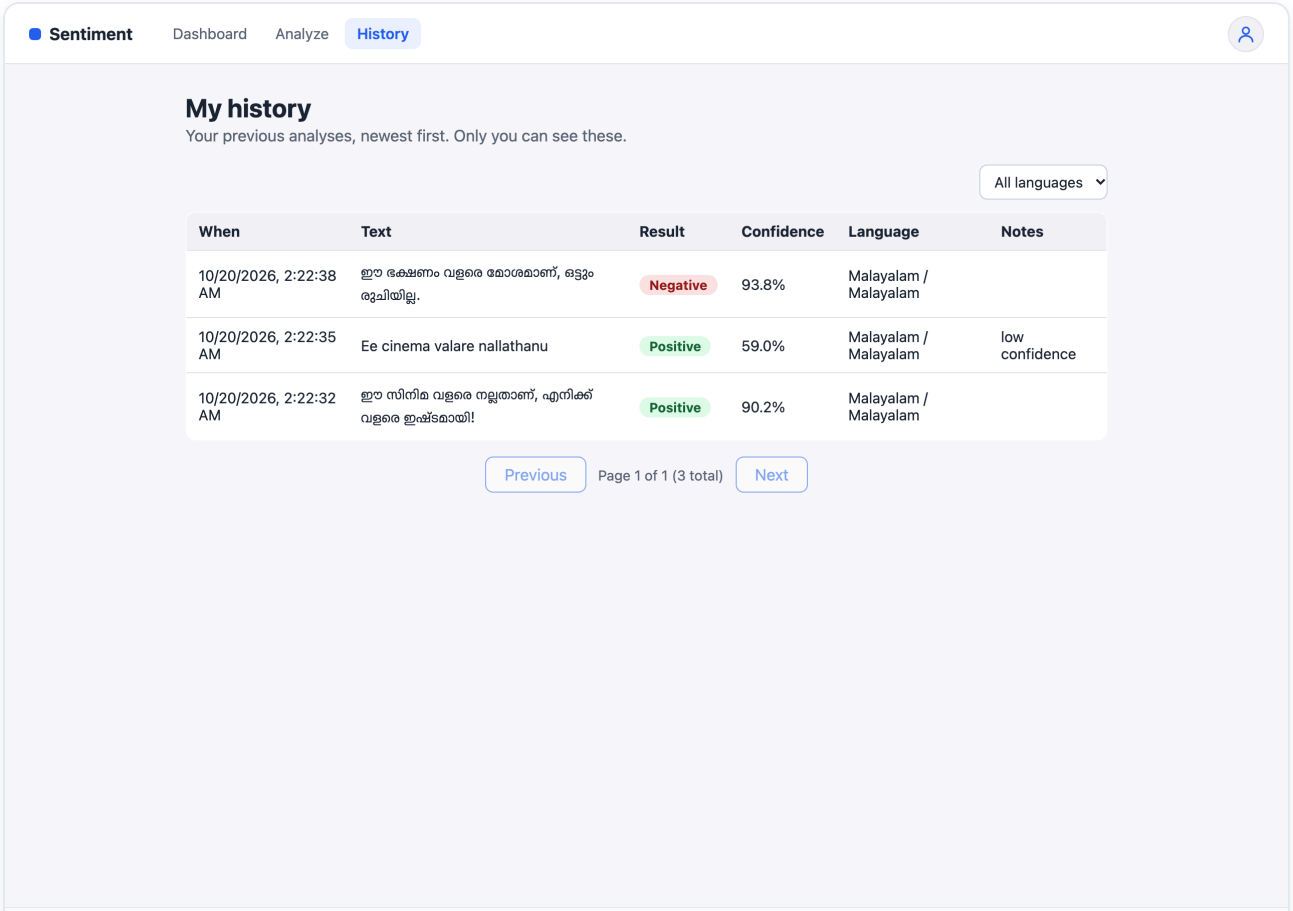
9. Understanding Results

After analysis, the result panel displays several pieces of information:

ELEMENT	DESCRIPTION
Sentiment Label	The predicted sentiment: Positive, Negative, or Neutral
Confidence Score	How confident the model is in its prediction (0%–100%). Higher is more reliable.
Probability Bars	Visual breakdown showing the probability for each of the three sentiment classes. The longest bar corresponds to the predicted label.
Language & Script	The detected (or selected) language and script type, e.g., "Malayalam (Malayalam script)" or "Malayalam (Latin script)".
Transliterated Text	Shown only for romanized input — the native-script version the model actually analyzed (blue box).
Latency	Time taken for the model inference in milliseconds.
Code-mixed (warning)	Appears when the text contains multiple scripts (e.g., Malayalam mixed with English).

ELEMENT	DESCRIPTION
Low confidence (warning)	Appears when the confidence is below the threshold (default: 60%). Treat the result with caution.

10. Analysis History



History page showing past analyses with pagination — `/user/history`

The History page shows all your past analyses in a table, sorted newest first. Key features:

Filtering by Language

Use the dropdown at the top to filter by a specific language:

FILTER OPTION	SHOWS
All languages	Every analysis you've run

FILTER OPTION	SHOWS
Tamil / Malayalam / Kannada / Telugu	Only analyses in the selected language
Not identified	Analyses where the language could not be detected

Pagination

Results are displayed 10 per page. Use the **Previous** and **Next** buttons at the bottom to navigate between pages. The current page indicator (e.g., "Page 1 of 1 (3 total)") is shown in the center.

✓ Your history is **private** — only you can see your own analyses. Other users cannot view your records.

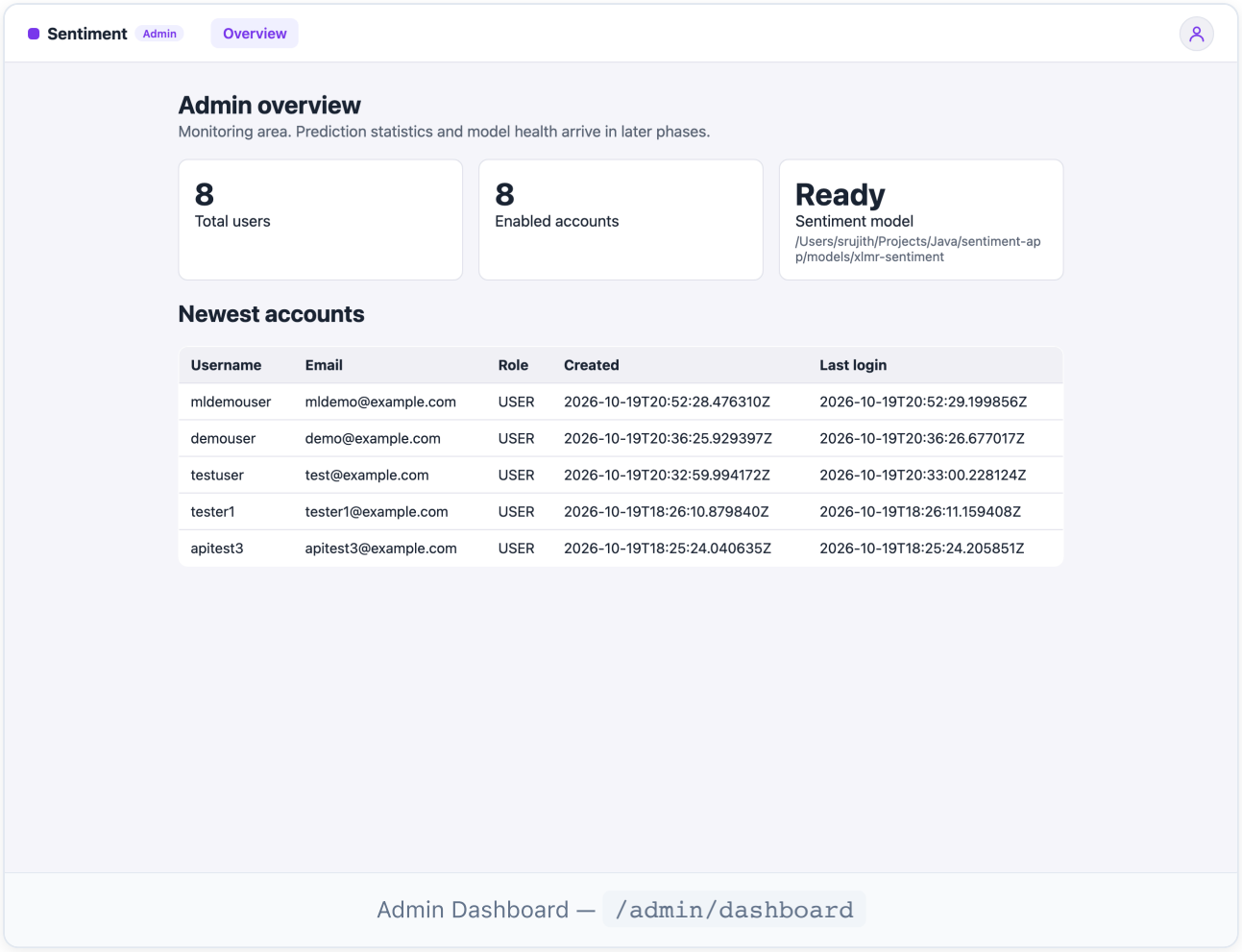
11. Supported Languages

LANGUAGE	CODE	NATIVE SCRIPT	ROMANIZED (LATIN)	TRANSLITERATION
Tamil	`ta`	தமிழ் (Tamil script)	"Indha padam romba nalla irukku"	Latin → Tamil
Malayalam	`ml`	മലയാളം (Malayalam script)	"Ee cinema valare nallathanu"	Latin → Malayalam
Kannada	`kn`	ಕನ್ನಡ (Kannada script)	"Ee cinema tumba chennagide"	Latin → Kannada
Telugu	`te`	తెలుగు (Telugu script)	"Ee cinema chala bagundi"	Latin → Telugu

ⓘ **Auto-detect** works best with native script. For romanized text, always select the language manually to enable transliteration and get accurate results.

12. Admin Dashboard

The Admin Dashboard is accessible only to users with the **ADMIN** role. Regular users cannot access this area.



Admin Dashboard — `/admin/dashboard`

The Admin Dashboard shows:

- **Total users** — number of registered accounts
- **Enabled accounts** — number of active (non-disabled) accounts
- **Sentiment model status** — whether the ONNX model is loaded and ready, along with its file path
- **Newest accounts table** — recent registrations with username, email, role, creation date, and last login

13. Keyboard Shortcuts & Tips

SHORTCUT / TIP	DESCRIPTION
`Ctrl` + `Enter` (or `⌘` + `Enter` on Mac)	Run analysis from the text area (same as clicking Analyze)

SHORTCUT / TIP	DESCRIPTION
Example chips	Click any chip below the text area to auto-fill sample text
Language memory	The language dropdown remembers your last selection across visits
Character limit	Maximum 1,000 characters per analysis. The counter shows remaining capacity.

14. Troubleshooting

ISSUE	SOLUTION
"Please contact the administrator" on the Analyze page	The sentiment model is not loaded. Contact your admin to check the model files at the configured path (`./models/xlmr-sentiment`).
"Invalid username or password" at login	Double-check your username and password. Ensure your account hasn't been disabled by an admin.
Low confidence / wrong result on romanized text	Make sure you **selected the correct language** from the dropdown. Without it, transliteration won't activate and accuracy will be low (~20%).
"Unknown language" error	You provided an unsupported language code. The system supports only Tamil (ta), Malayalam (ml), Kannada (kn), and Telugu (te).
"Please enter some text" error	The text area is empty or contains only whitespace. Enter actual text before clicking Analyze.
Cannot access `/admin/dashboard`	Only ADMIN-role accounts can access admin pages. Regular user accounts will see a 403 Forbidden error.
Page not loading / connection refused	Ensure the application is running and accessible at the correct URL (default: `http://localhost:8080`).

Sentiment Analysis Application — Cross-lingual Dravidian Language Sentiment Analyzer

Model: cardiffnlp/twitter-xlm-roberta-base-sentiment • Spring Boot 3.5 • Java 21 • DJL / ONNX Runtime • ICU4J 75.1